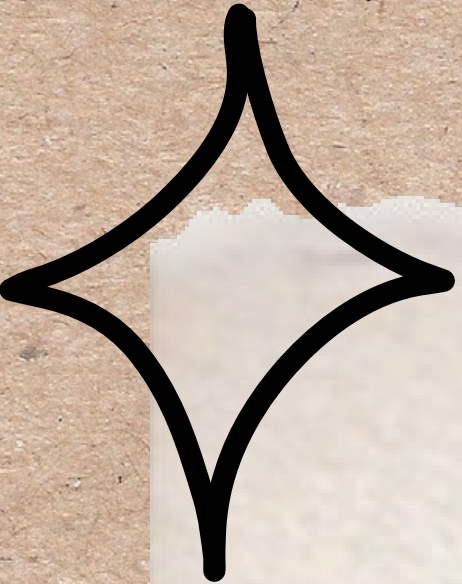


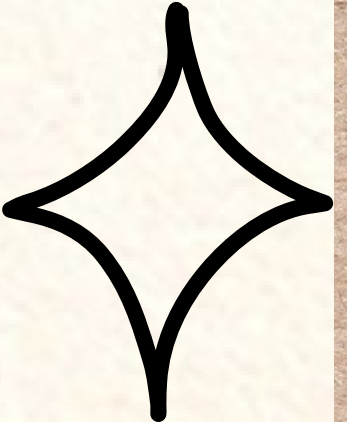
A low-angle, upward-looking photograph of several modern skyscrapers with glass facades. The buildings are reflected in each other's windows, creating a complex geometric pattern. The sky is a deep blue with wispy white clouds. The overall tone is professional and tech-oriented.

EXPLORING CLICKSTREAM DATA

A guide through useful business
insights for an e-commerce firm

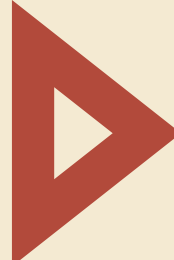


Presentation Plan

- Business Significance
 - Problem Scenario
 - Data Involved and Brief Overview
 - Steps involved in the Analysis
 - Hypothesis Testing b/w different types of reviews
 - Determining Customer Journey
 - Recommendations
- 



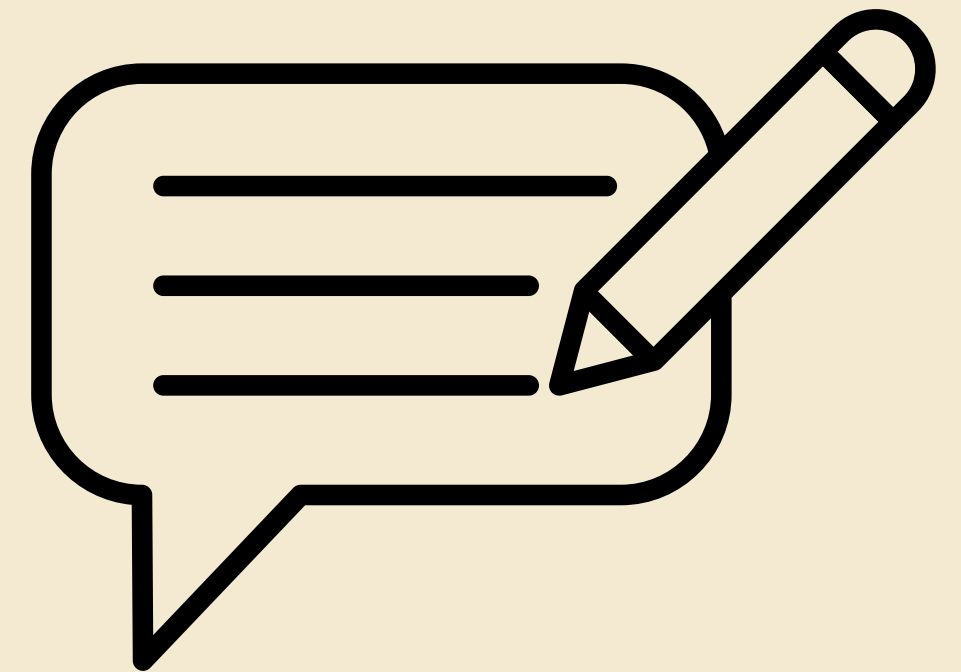
Business Significance



Understanding
User Behaviour

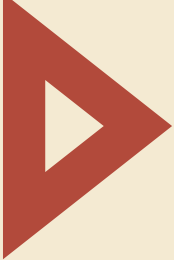


Personalisation
and Targeting



Conversion
Optimisation





Problem Scenario

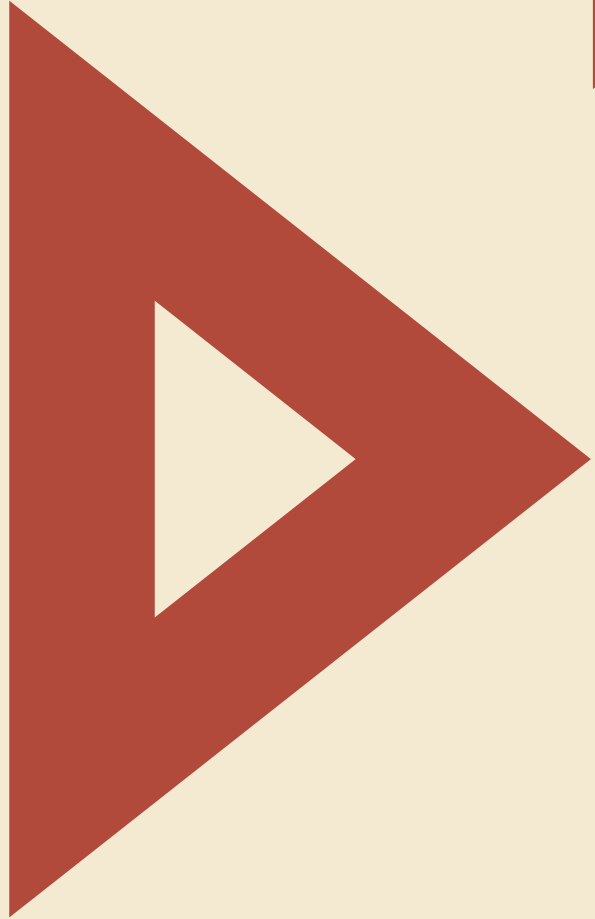
Defining the objective

To understand how customers behave and what drives their choices online



What key aspects determine consumer behaviour ?

- Influence of different sources of product reviews and which review type performs ensures high returns
- Mapping customer journey in terms of most commonly visited pages, buying trends in lieu of gender, the average number of visits etc.



How to effectively use the inferences gained?

Recommend strategies to optimise the user experience, product performance and overall conversion rates

Data Information



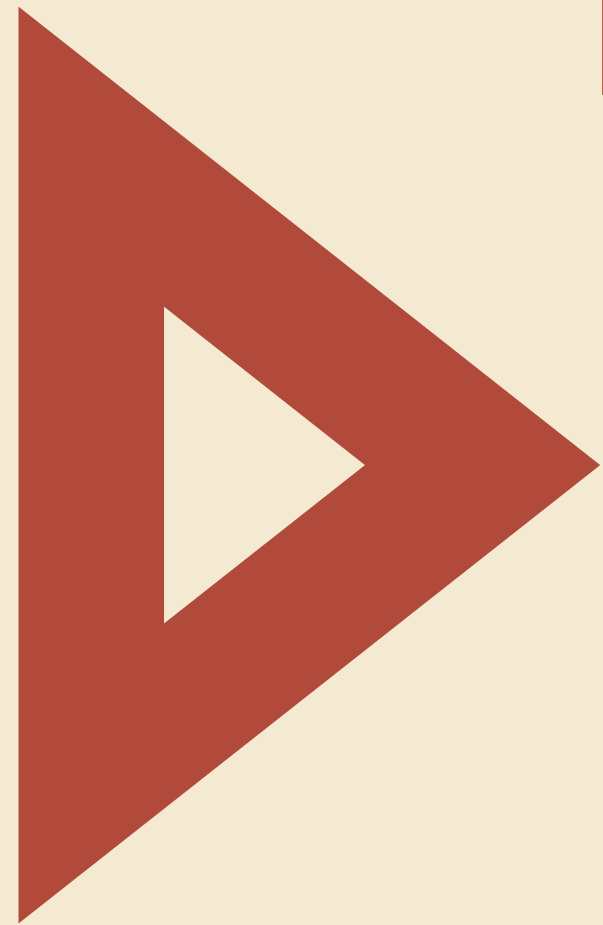
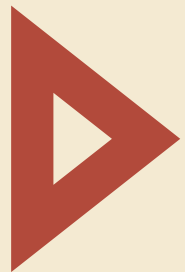
CLICKSTREAM FEED
DATA



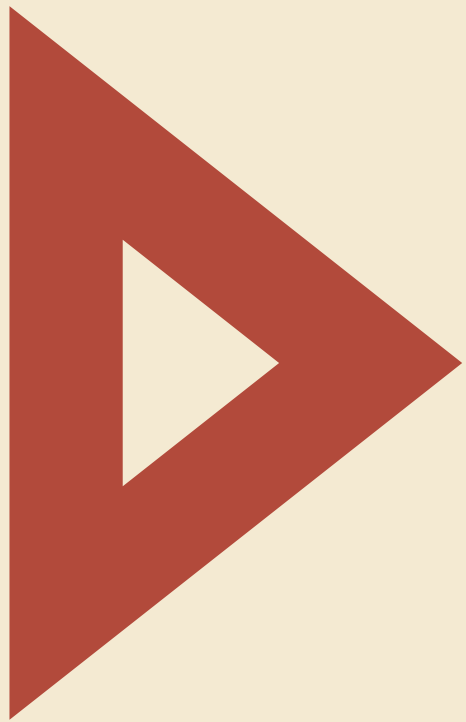
REVIEWS
AND PRODUCTS



REGISTERED USERS



Brief Overview



Review Types



Customer Review

Written feedback revolving around product quality and price points

Video Review

Recorded reactions enabling potential buyers to enjoy the feel of the product & make right decisions

Celebrity Recommendation

Endorsements made by influencers providing product's credibility and brand's visibility.

Methodology



Data Preprocessing



Statistical Analysis



Visualisation



Handling the data

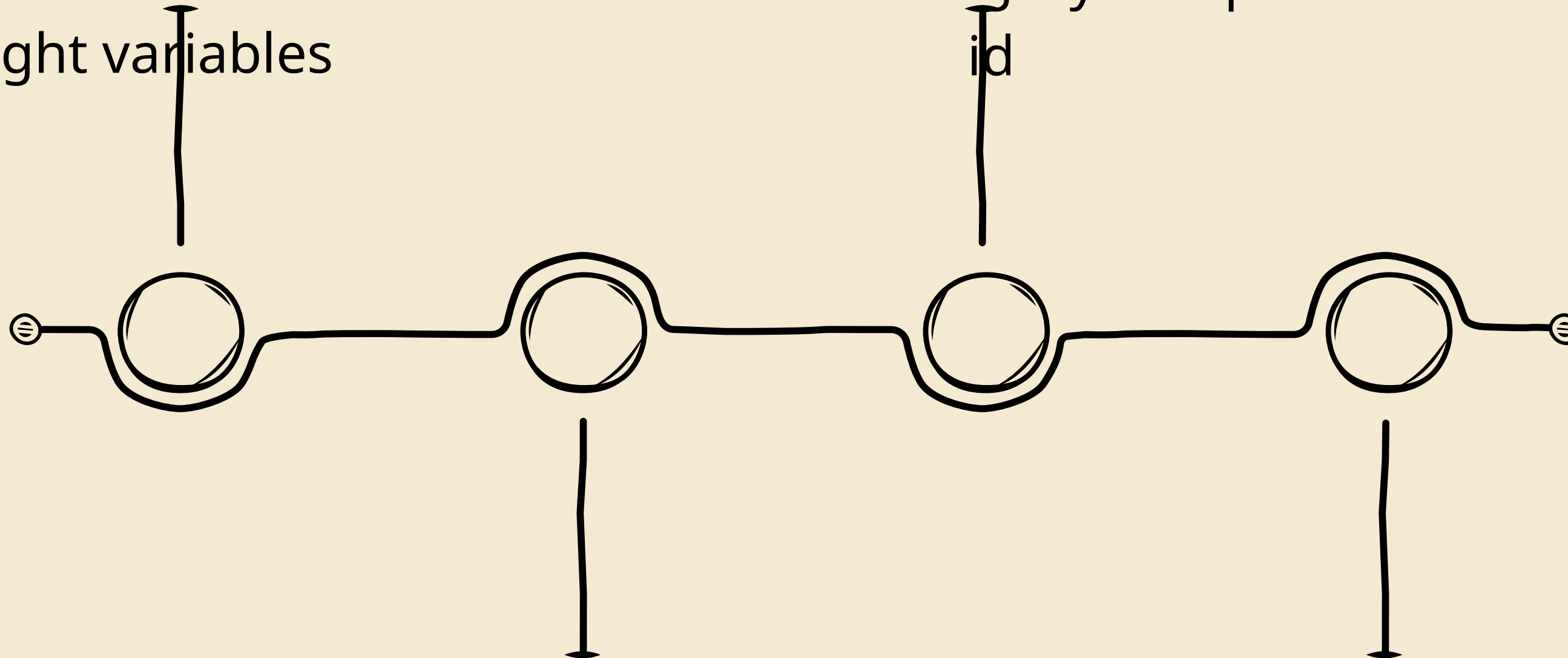
Data Cleaning

Segregated the data
points
under the right variables

Data Aggregation

Grouped the data points based on
each review category and product

id



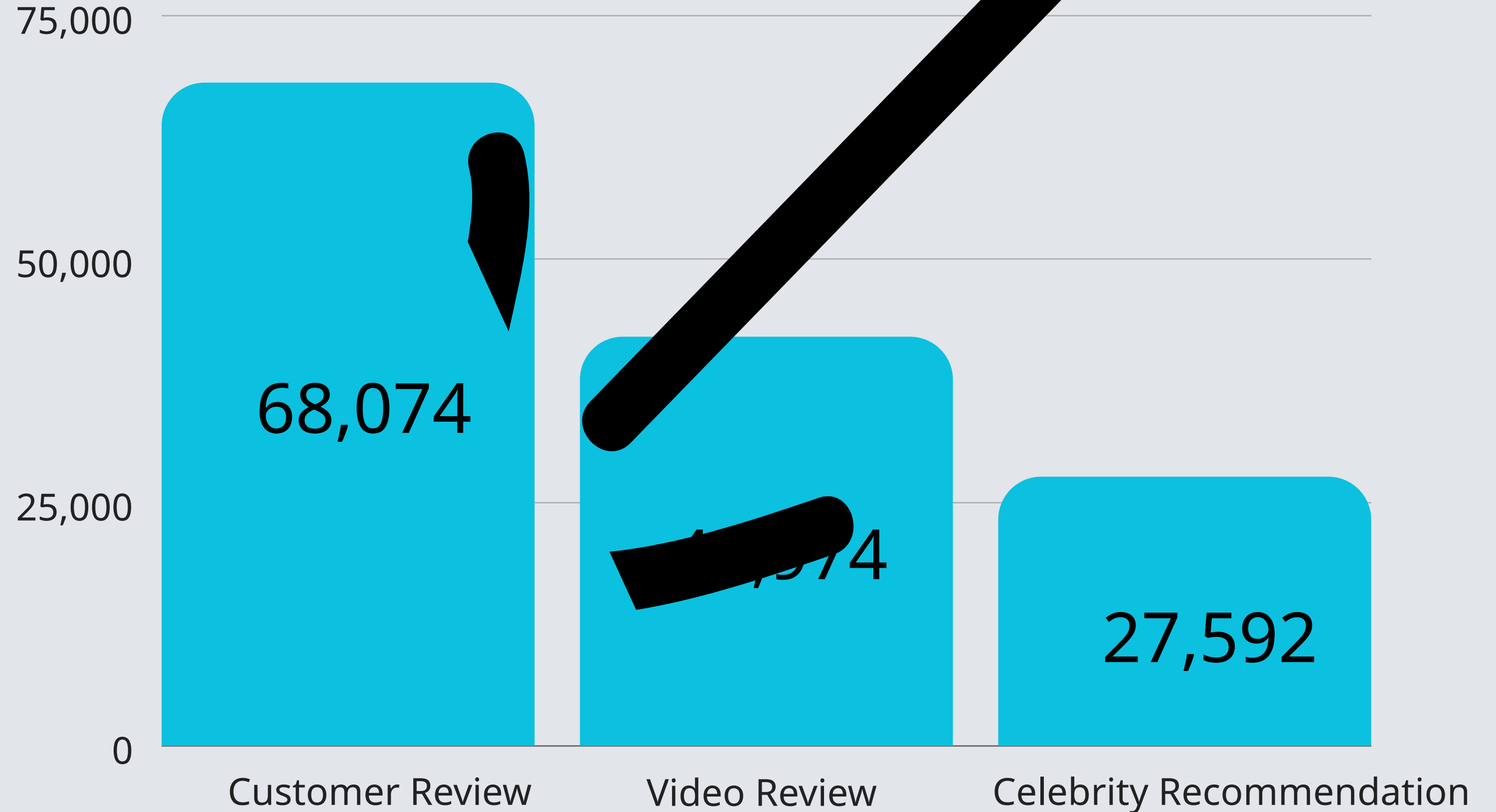
Data Transformation

Classified each session under
the

Data Analysis

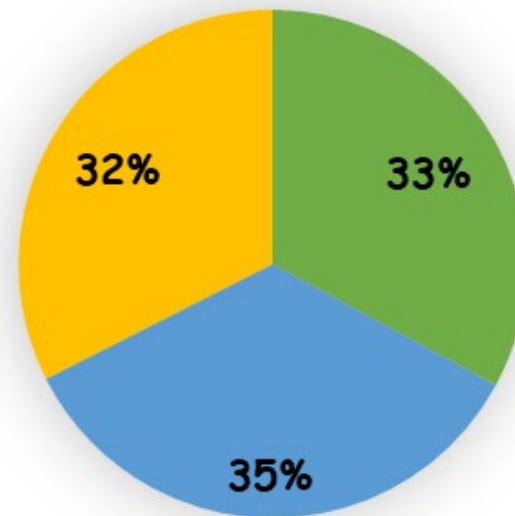
Aggregated the count of session
ids

Sessionwise Distribution of Reviews



Productwise Distribution of Reviews

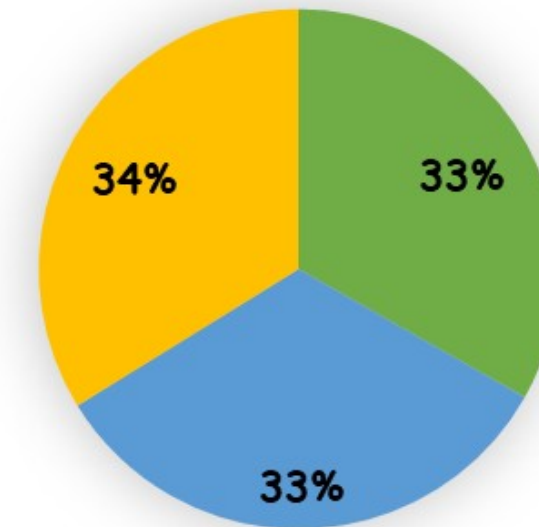
Product 1



■ Customer Review ■ Video Review ■ Celebrity Recommendation

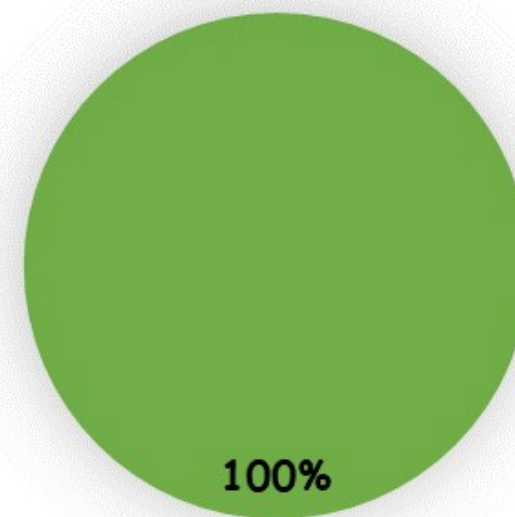


Product 2



■ Customer Review ■ Video Review ■ Celebrity Recommendation

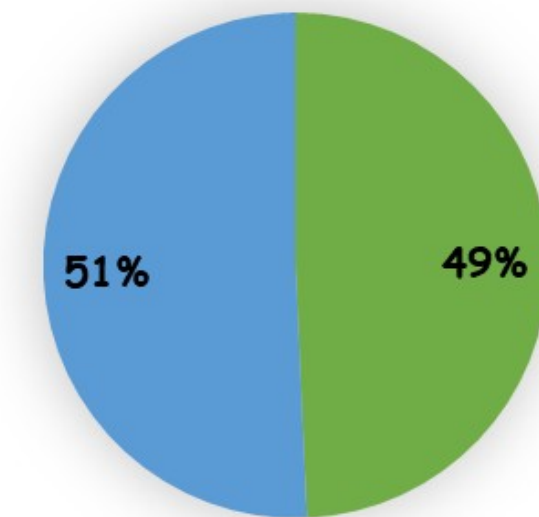
Products 4 and 5



■ Customer Review



Product 3



■ Customer Review ■ Video Review

Are celebrity reviews more influential than customer reviews ?

Proportions



Customer Reviews = 0.26



Video Reviews = 0.16



Celebrity reviews = 0.11

Hypothesis Test

-145.03

Z-score for the
difference in
sample proportions

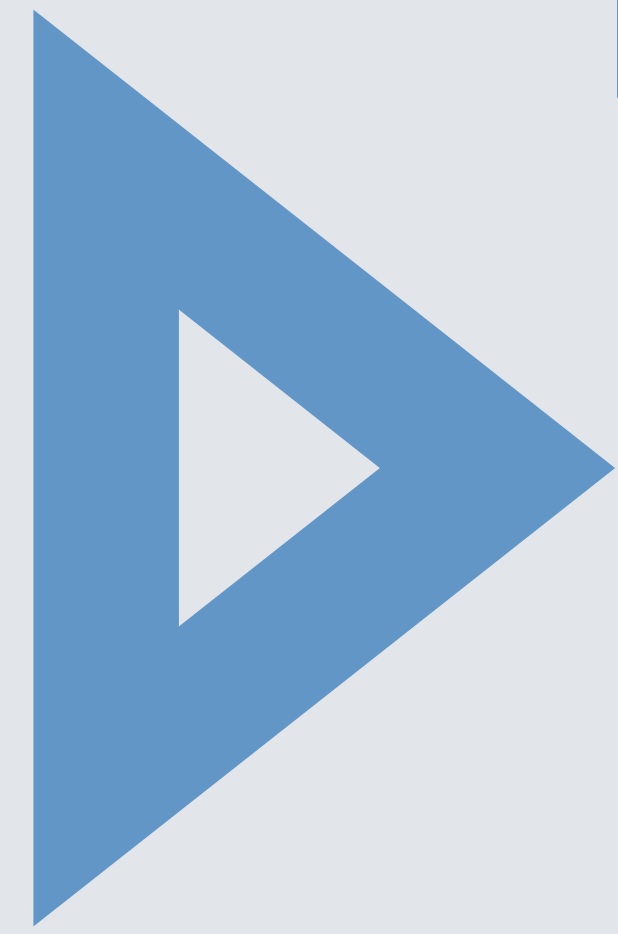
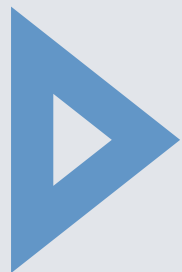
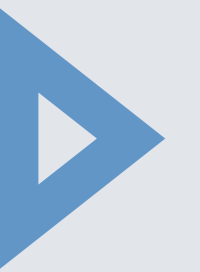
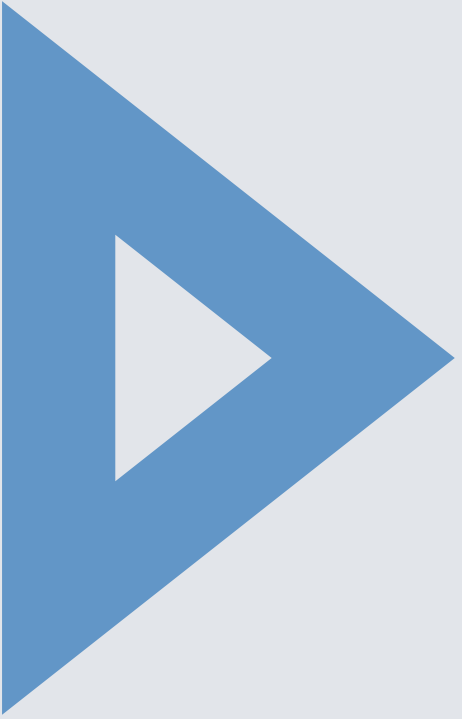
Corresponding right-tailed
p-value

≈ 1

Since we fail to reject the null hypothesis at 5% level, the following conclusion is drawn:

Celebrity reviews do not prove to be more influential than customer reviews.

How celebrity and customer
reviews fare against video
reviews ?



Celebrity v/s Video

Customer v/s Video

-58.63

Z-SCORE FOR THE
DIFFERENCE IN
SAMPLE PROPORTIONS

88.7

≈ 1

RIGHT-TAILED P-VALUE

≈ 0

Celebrity reviews do
not perform better
than video reviews

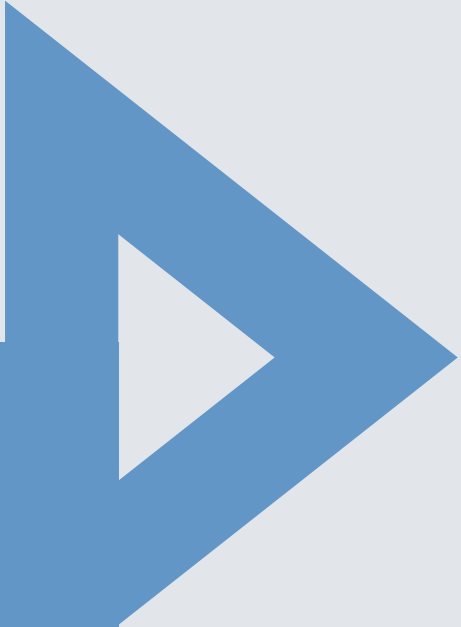
CONCLUSION

Customer reviews do
perform better
than video reviews

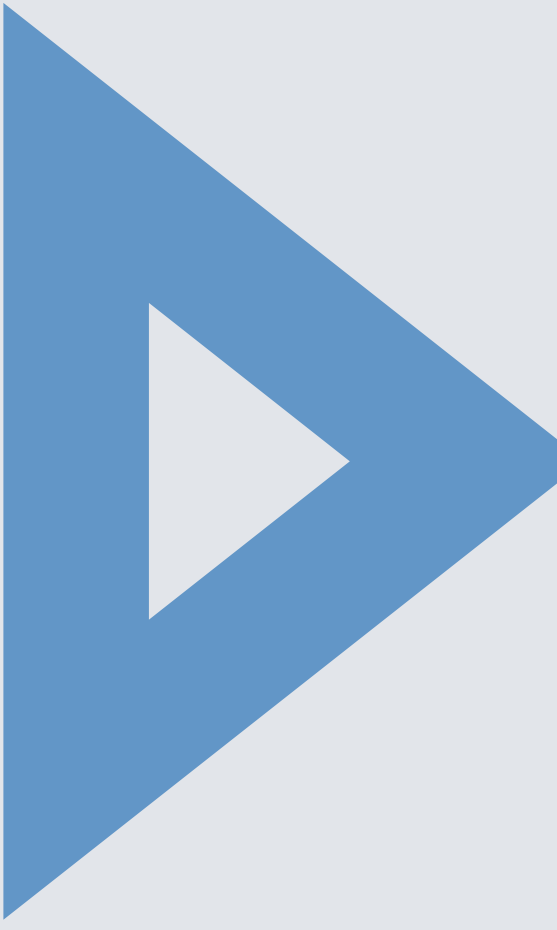
Capturing the "bump"

$$\text{Total Purchases} = \beta_1 \text{ customer review} + \beta_2 \text{ celebrity review*video review}$$

Variables	Coefficient
CUSTOMER REVIEW	S 0.1814
CELEBRITY + VIDEO REVIEWS	0.003761

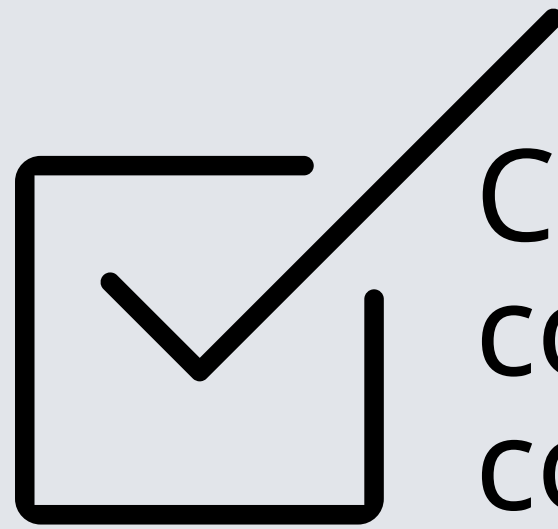


Measures and policies need to be adopted to improve the performance of influencer ratings and video reviews.



Customer feedback posts should be promoted to boost sales.

Summarising the solution



Celebrity reviews are least influential in comparison to the other reviews when considered pairwise.



Customer reviews generate more income for the business than celebrity endorsements and video reviews put together.

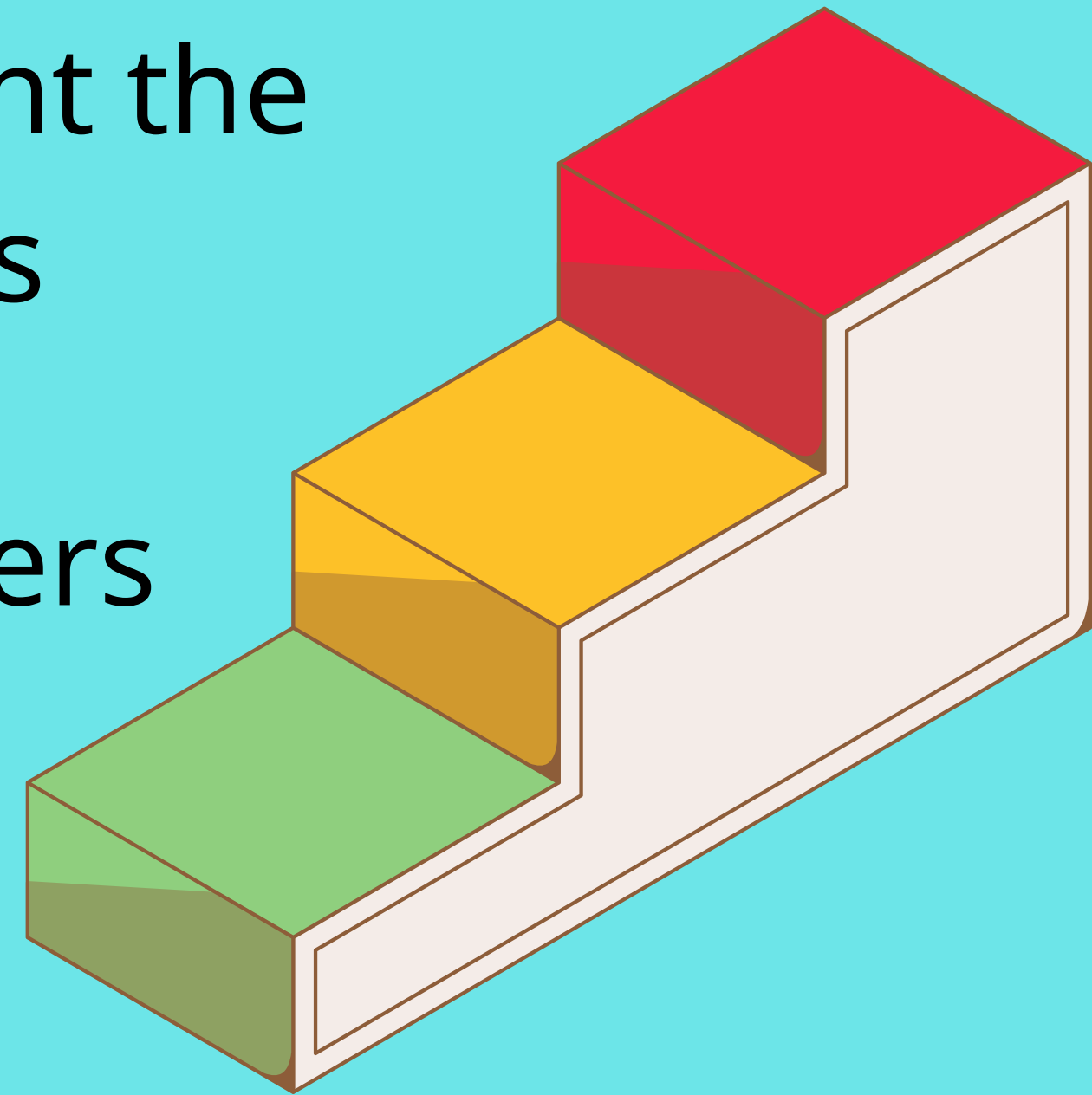
DETERMINING CUSTOMER JOURNEY

- Top 6 most visited pages
- Path followed by a set of random users and their exit pages
- Average number of pages visited by buyers vs. non-buyers
- Average number of pages visited by males vs. females

Key
Indicators

A walk through the steps.....

- 1 Sort the data by user id and timestamp
- 2 Group the data by url and count the number of distinct occurrences
- 3 Filter out a random set of 5 users
- 4 Apply the aggregate function on the time series object



Top Most Website Landing Pages

1

<http://www.RL.com/>
(homepage)

4

[http://www.RL.com/
product/4005](http://www.RL.com/product/4005)

2

[http://www.RL.com/
product/4003](http://www.RL.com/product/4003)

5

[http://www.RL.com/
product/4002](http://www.RL.com/product/4002)

3

[http://www.RL.com/
product/4001](http://www.RL.com/product/4001)

6

[http://www.RL.com/product/
4004](http://www.RL.com/product/4004)

Sequence of Pages Visited by Random Users

review/3002 → homepage → product/4002

exit

User ID
26B45553-8BE2-4AB0-ABE5-ACDB78B76410

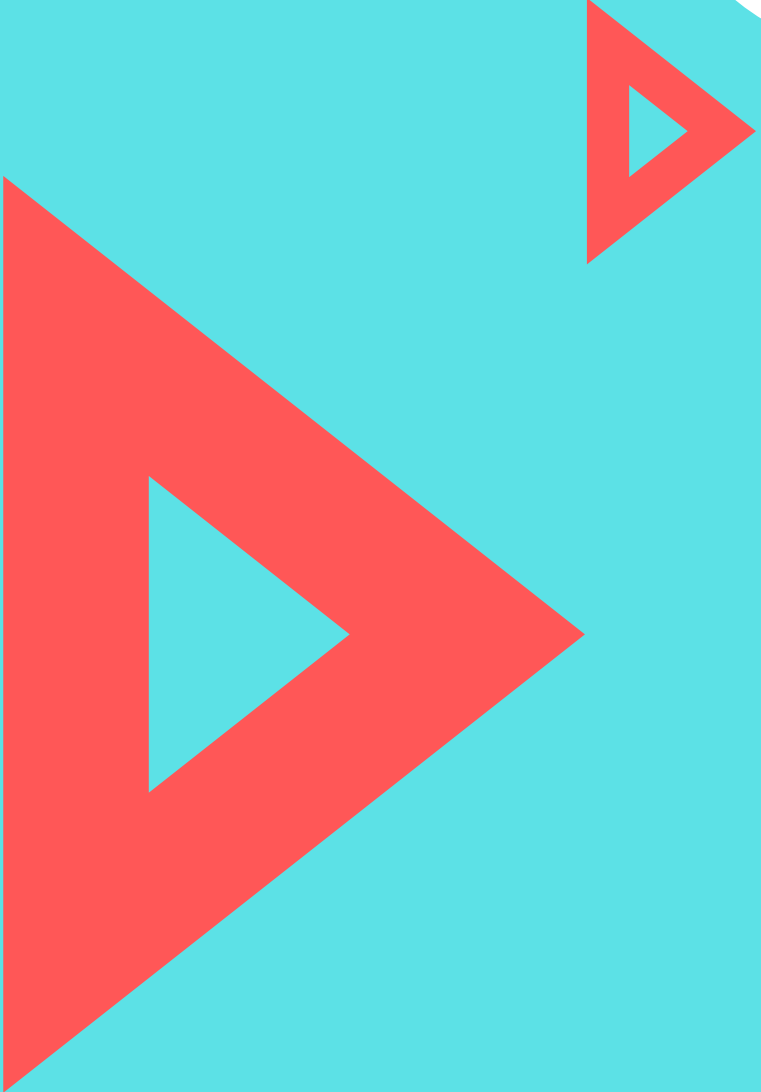
product/4004

video/3002

review/3002



AVERAGE
NUMBER OF
PAGES VISITED



$1.652 \approx 2$

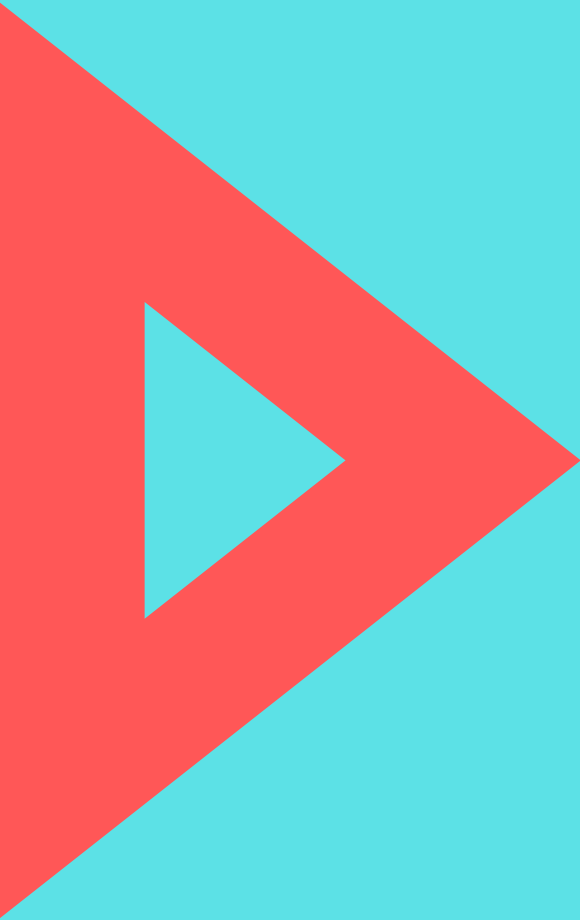
Buyers

$8.51 \approx 9$

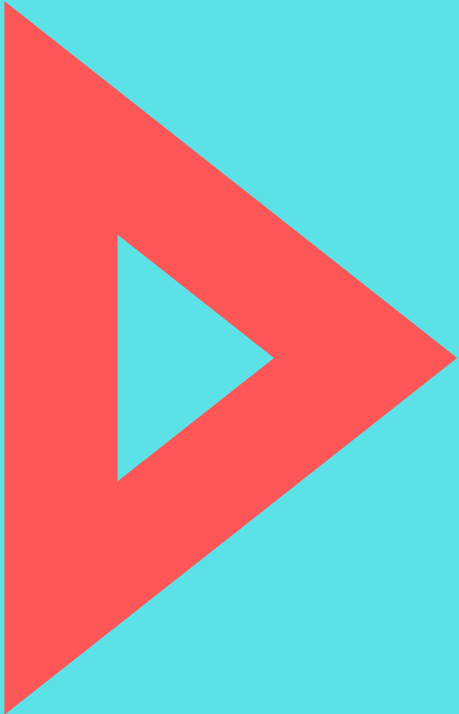
Non-Buyers



AVERAGE
NUMBER OF
PAGES VISITED



10.37 $\approx$ 1
8 Males 0



10.72 $\approx$ 1
7 Females 1

CONCLUSION AND RECOMMENDATIONS

- INVESTIGATE USER EXPERIENCE
 - ASSESS THE IMPACT OF REVIEWS ON REVENUE
- VISUALIZE CUSTOMER JOURNEYS FOR ENHANCED PERFORMANCE
 - EMPLOY A/B TESTING TECHNIQUE TO IDENTIFY HIGH POINTS
- DEVISE A FINANCIAL PLAN TO ALLOCATE RESOURCES EFFECTIVELY
- IMPROVISE THE FEEDBACK PLATFORM AND KEEP REGULAR CHECKS
- ADOPT MARKETING CAMPAIGNS TO INCREASE CUSTOMER RETENTION
RATE

THANK YOU!!